

HOTEL BOOKING & RESERVATION ANALYSIS

Project Brief • Dashboard Story • Insights & Business Decisions

1. Problem Statement

Hotels generate a large volume of reservation data, but booking volume alone does not explain business performance. Management needs to understand where bookings come from, how reliable those bookings are, how pricing behaves, how long customers stay, and which customer, room, meal and geographic patterns contribute to booking value.

The core problem is therefore to transform reservation-level data into clear, interactive information that can help management identify demand patterns, cancellation risks and value-driving segments, and then support practical business decisions.

2. What Data We Have

The project uses hotel reservation records containing information about the hotel, booking timing, customer and market characteristics, room information, stay duration, pricing and reservation status.

Key analytical fields include:

- Hotel type — City Hotel / Resort Hotel
- City / geographical location
- Arrival month and booking timing
- Market segment — Online TA, Offline TA/TO, Groups, Corporate, Direct, Complementary and others
- Customer type — Transient, Transient-Party, Contract and Group
- Deposit type
- Meal arrangement
- Reserved room type
- Stay duration — weekday and weekend nights
- ADR — Average Daily Rate
- Reservation status / cancellation status

These fields allow the dashboard to move from simple booking counts to deeper questions about customer behaviour, pricing and booking value.

3. What We Want to Show

The dashboard is designed around four connected questions:

1. DEMAND — How many bookings are coming in, when do they occur, and where are they concentrated?
2. STABILITY — How much of that demand is canceled, and which hotel, customer or market segments show greater cancellation risk?
3. VALUE — What pricing and stay-duration patterns are associated with the bookings?
4. CUSTOMER & PROPERTY BEHAVIOUR — Which customer types, market segments, room types, meal choices and cities are associated with different booking patterns?

The aim is not to show charts for the sake of showing charts. Each visual should answer a business question.

4. Dashboard Story — Page 1: Booking Performance

Provides an executive view of booking volume, cancellation and booking behaviour.

KPIs:

- Total Bookings — overall reservation volume.
- Cancellation Rate — proportion of bookings cancelled; a measure of demand stability.
- Checked-Out Bookings — bookings reaching the check-out status in the dashboard calculation.
- Average ADR — average daily room rate; a pricing indicator, not profit.
- Average Lead Time — how far in advance customers make reservations.

Filters such as Hotel, City, Arrival Month, Market Segment, Customer Type and Deposit Type allow management to move from the overall picture to a specific business segment.

Key visuals:

- Monthly booking trend — identifies changes in demand over time.
- Top cities by bookings — identifies where booking volume is concentrated.
- Bookings vs cancellations by hotel — compares City Hotel and Resort Hotel behaviour.
- ADR by city — compares pricing levels geographically.
- Cancellation rate by market segment — identifies segments requiring closer investigation.

5. Dashboard Story — Page 2: Revenue & Customer Analysis

It moves from reservation volume into booking value, customer composition and product preferences.

KPIs:

- Total Bookings — volume baseline.
- Average ADR — daily pricing level.
- Average Stay Nights — average duration of a stay.
- Estimated Stay Value — conceptually $ADR \times \text{stay nights}$; an estimated room-value measure, not profit.
- Revenue Per Booking — average value at booking level according to the dashboard calculation.

Filters:

- Hotel and City — property and geography.
- Meal — meal arrangement associated with the booking.
- Reserved Room Type — room category originally requested.
- Market Segment — source/category of demand.
- Customer Type — nature of the booking/customer.
- Deposit Type — booking commitment/payment condition.

Key visuals:

- Revenue by customer type — shows where the displayed value is concentrated across customer categories.
- Average ADR by month — shows how pricing changes through time.
- Guest composition by hotel type — compares guest volume associated with City and Resort Hotels.
- Stay Length vs ADR scatter plot — examines the relationship between stay duration and daily pricing.
- Revenue contribution by city — shows geographical distribution of the revenue-related measure.

6. Core Insights to Conclude

1. High booking volume does not automatically mean successful or valuable business. Booking volume should be read together with cancellation and value measures.
2. Cancellation is not evenly distributed across segments. High-risk segments should be investigated separately rather than applying one policy to every booking.
3. City Hotel and Resort Hotel should not be treated as identical businesses. Their booking and cancellation patterns can differ substantially.
4. Booking volume and monetary value are different perspectives. A high-volume city or segment is not automatically the highest-value one.
5. ADR must be interpreted together with stay duration. A higher daily rate does not automatically produce a higher total booking value, and neither ADR nor stay duration alone proves profitability.
6. Customer type, market segment, room type and meal choice help explain where booking behaviour and value are coming from.
7. Geography matters. City-level analysis can reveal different combinations of booking volume, ADR, cancellation and value, supporting more targeted decisions.

The Data Checkmates